Name : **Class : B.E.** **Roll No :**

PRACTICAL NO. 04

K-Means Clustering on the Iris Dataset using the Elbow Method

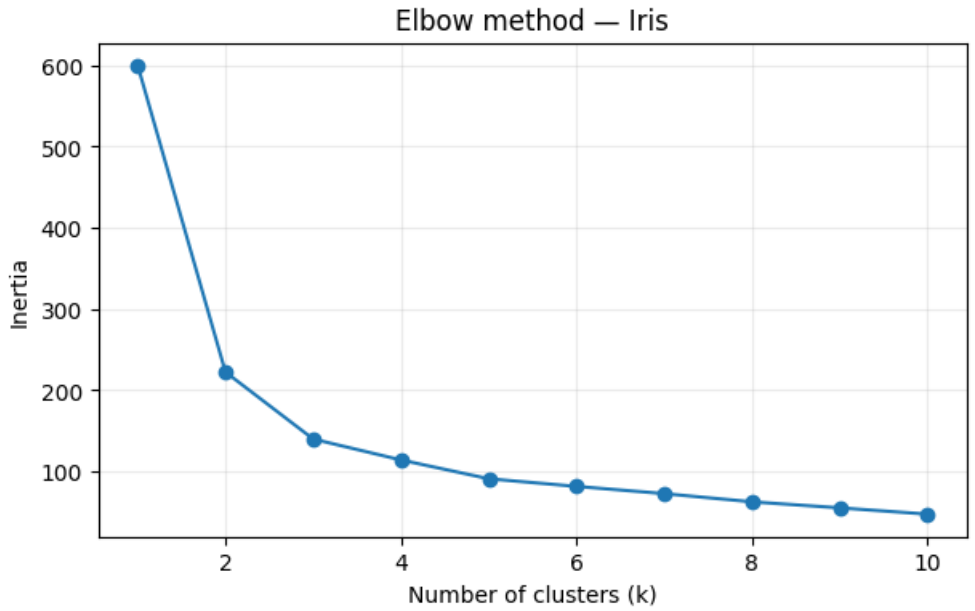
Aim

To implement K-Means clustering on the Iris dataset, use the elbow method on within-cluster inertia to choose the number of clusters k, and evaluate the final clustering with cluster counts and the silhouette score.

Code and Output

In [1]:
```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
```

In [2]:
```
iris=load_iris()
X=StandardScaler().fit_transform(iris.data)
k_values=range(1,11); inertias=[]
for k in k_values:
    model=KMeans(n_clusters=k,random_state=42,n_init=10).fit(X)
    inertias.append(model.inertia_)
plt.figure(figsize=(7,4)); plt.plot(list(k_values),inertias,marker="o")
plt.xlabel("Number of clusters (k)"); plt.ylabel("Inertia")
plt.title("Elbow method - Iris")
plt.grid(alpha=.25); plt.show()
print("Inertia values:",[round(v,2) for v in inertias])
```

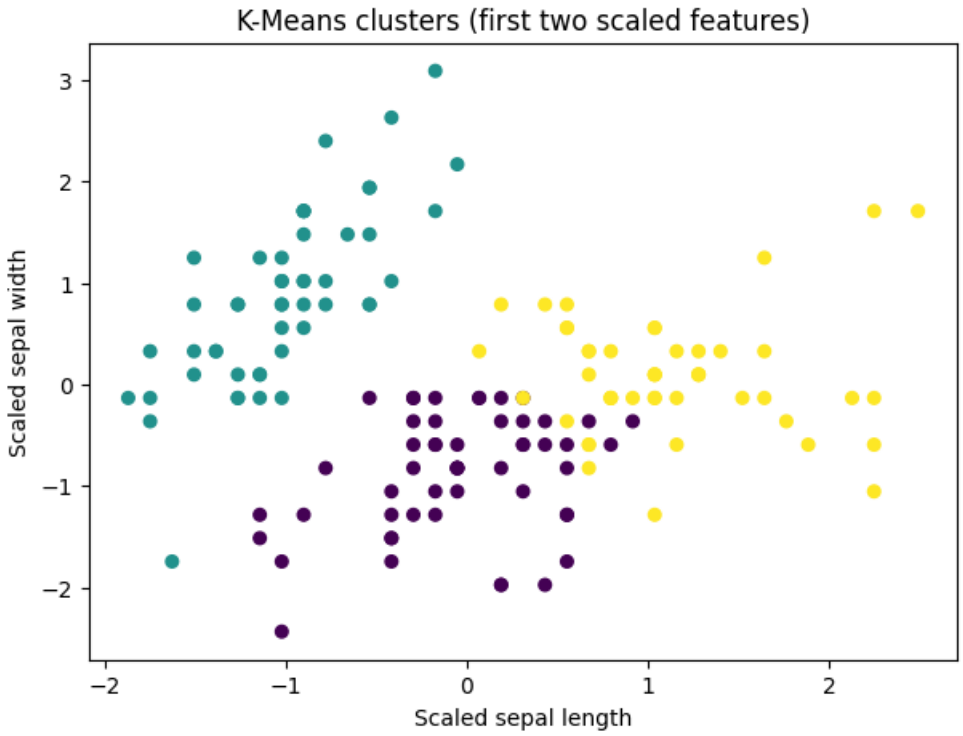



```
Inertia values: [600.0, 222.36, 139.82, 114.09, 90.93, 81.54, 72.63, 62.54, 55.12, 47.39]
```

In [3]:

```python
# Iris has three known species; choose k=3 for the final clustering.
k=3
model=KMeans(n_clusters=k,random_state=42,n_init=10)
labels=model.fit_predict(X)
print("Chosen k:",k)
print("Cluster counts:",{int(i):int((labels==i).sum()) for i in set(labels)})
print("Silhouette score:",round(silhouette_score(X,labels),4))
plt.figure(figsize=(7,5))
plt.scatter(X[:,0],X[:,1],c=labels,cmap="viridis",s=30)
plt.xlabel("Scaled sepal length"); plt.ylabel("Scaled sepal width")
plt.title("K-Means clusters (first two scaled features)"); plt.show()
```

```
Chosen k: 3
Cluster counts: {0: 53, 1: 50, 2: 47}
Silhouette score: 0.4599
```


K-Means clusters (first two scaled features)
Scaled sepal width
Scaled sepal length
3
2
1
0
−1
−2
−2
−1
0
1
2